Observer Patch Holography as a Strange-Loop Self-Simulation
The Strongest Results and Their Certificates

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Observer-Patch Holography (OPH) is a theory-of-everything candidate with zero continuous dials, closure-defined constants, machine-certified uniqueness of its fixed points, a structural core carried by 111 sorry-free Lean theorems, a standing falsification program with cryptographically frozen and externally timestamped targets, and a survived hostile 42-finding third-party audit that identified no false theorem in the recovered core. That combination exists nowhere else among theory-of-everything programs, and every item on the list is a checkable public artifact.

OPH rests on a single assumption: reality is a self-referential mathematical structure that explains itself, with no external machine and no external clock. Everything else follows from consistency. The architecture and settings of the simulation, the existence of observers, and the emergent universe they see are forced by consistency conditions. Each observer is a bounded self-reading patch with local state, ports, boundary readback, durable records, mismatch feedback, repair moves, and public receipts, and no patch contains the world. The decisive demand is that observers agree on the reality that emerges, and that requirement of mutual agreement pins down our exact universe, the stable public fixed point that survives when all local views are made mutually consistent,

$$T(\mathfrak{U}_{\mathrm{OPH}}) = \mathfrak{U}_{\mathrm{OPH}}.$$

Douglas Adams proposed that the Earth is a computer running a very long calculation, and he was joking. OPH makes the adjacent claim with the joke removed: the universe is a computation checking its own records, and the output is on public display. Take a structure that has to explain itself, let consistency force finitely many observers that keep records and argue until their notes agree, model the argument as a fixed-point computation on a holographic screen, and solve. Out falls a universe whose events obey Born-rule probabilities on a central record algebra [O1,O3], whose observers see $3+1$-dimensional Lorentzian spacetime running Einstein's equations on the declared branches [O4,C8], whose horizon closes at the de Sitter capacity $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$ so the sky reads as an accelerating expansion with a small cosmological constant [O1,O4], and whose particle catalogue is the Standard Model quotient with three colors, three generations, and one Higgs doublet [O4]. The full answer runs to rather more decimals than 42. None of it is fitted and no dial is turned. A construction forbidden from touching the data reproduces the signature of spacetime, the gauge group, the generation count, and the fine-structure constant; if self-consistency were not the reason, that agreement would be the most expensive coincidence ever tabulated.

The argument, and the price of every exit

This document runs one assent chain. **Step 1:** grant the six strange-loop principles below [C17]. **Step 2:** verify the certificates; the interval uniqueness certificate for the pixel closure [C1], the conditional capacity fixed point [C16], and the 111 sorry-free Lean theorems [C6] are machine-checkable objects, and a hostile third-party audit reproduced the certificate arithmetic end to end. **Step 3:** observe the landings in the results table, each carrying its epistemic label. What survives as freedom after these three steps is the enumerated open program: the Ward-projected hadronic transport term, the electroweak physical readout contract, and the capacity readback map F with its premises, each with a declared certificate schema and a registered kill condition. Every exit from the chain costs a named, checkable object: striking a principle names the observation that justifies the strike, rejecting a certificate names an arithmetic error in a public artifact, and disputing a landing names the ledger row whose label is wrong.

The strange-loop principles [C17]

Every claim in this document is deduced from six declared strange-loop principles, condensed here from the canonical statement in the program's claim registry [C17]. **SL-0 (Strange loop):** reality is a self-referential mathematical structure that explains itself, the fixed point of a self-reading computation; the simulating structure and the simulated world are one closed system, with no external machine and no external clock. Everything else follows from consistency: the architecture, the observers, and the emergent universe are forced, and observer agreement on the emergent reality pins down our exact universe. **SL-1 (Substrate chart):** the observer-facing substrate is charted by the two-sphere with the screen-microphysics architecture (patches, records, overlap comparison, repair); $3+1$-dimensional observer experience is a design constraint of the substrate under SL-0. **SL-2 (Detuning law):** the pixel ratio sits off the golden-ratio balance point by the Gaussian-normalized width of one electromagnetic observation, $P = \varphi + \sqrt{\pi}\,\alpha$; a screen at exact balance carries no events. **SL-3 (Constant identity):** under self-simulation the fine-structure constant of the simulated world and the substrate pixel readout are one quantity; the theory computes P from its closure equation and the measured α is the test that value faces. **SL-4 (Capacity identity):** the total record capacity N is one number shared by substrate and world; the theory computes N from $N = F(N)$ and the Λ readout is the test that value faces. **SL-5 (SI bridge):** one clock anchor, the cesium transition, connects substrate units to laboratory units.

Zero dials

The theory layer takes no measured number: zero continuous fit parameters in every defining equation. Both constants are defined by closure equations,

$$P_\star = \Gamma_P(P_\star), \qquad N_\star = \Gamma_N(N_\star),$$

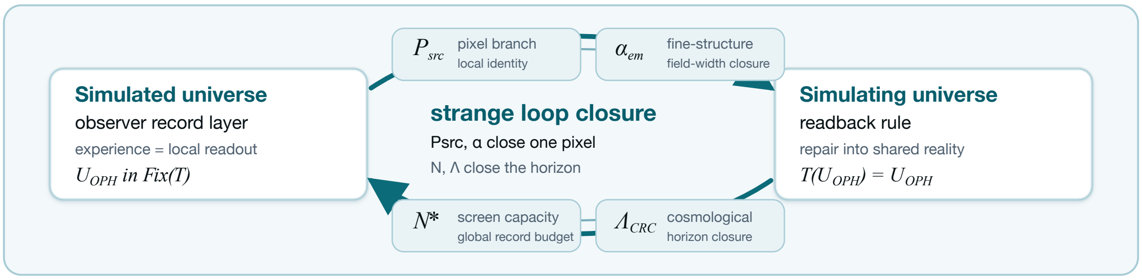
and nothing in the architecture can be adjusted toward data. Discrete structural selections are counted, with their menu sizes, in the public closure ledger [C17]. Measured values enter at the test layer as comparisons, and at the working layer as located values inside lanes whose closure terms are open: the working pixel $P = 1.630968209403959\ldots$ located from CODATA $\alpha^{-1} = 137.035999177(21)$ [S2], and the working capacity $N = 3.31 \times 10^{122}$ located from the Λ readout [S5]. Every working borrow is tied to a named generator and retires when that generator lands. The integers $(N_c, N_g, \beta_{\mathrm{EW}}) = (3, 3, 4)$ are branch outputs, and their alternatives are structurally excluded by a preregistered sweep.

Exactly one fixed point, certified

The pixel closure has exactly one fixed point per declared map on the declared physical domain $\alpha^{-1} \in [100, 200]$: existence and local uniqueness by interval Banach in outward-rounded arithmetic with contraction constant $L \leq 0.0724$, and global at-most-one by certified $\sup |g'| < 1$ on all 256 subdivision pieces with zero exceptional set [C1]. The enclosure width is 7.2×10^{-24} in α^{-1}, with the SU(2)/SU(3) edge-sum tails bounded by geometric majorants so the enclosure covers the infinite-cutoff sums. A hostile third-party audit reproduced the certificate arithmetic end to end. There is no landscape to relocate into: a second fixed point anywhere on the declared physical domain is arithmetically excluded.

The flagship numbers

The gauge-width declared map lands at $\alpha^{-1} = 137.035660$, a relative 2.5×10^{-6} from the measured $\alpha^{-1} = 137.035999177(21)$ [S2], a certified fixed point with the Ward-projected hadronic transport term open. The source map lands at 136.994835, a relative 3.0×10^{-4}, under the same label. The chart W coordinate sits 0.5 propagated experimental standard deviations from the PDG 2026 complex pole [S3], a convention diagnostic with the physical readout contract open. Read as $\Lambda \ell_P^2 = 3e^{-6\pi/(P\alpha_U)}$, the capacity bridge writes the 122-order vacuum-energy hierarchy as one exponential of the inverse gauge coupling, with zero continuous freedom once α is measured and no 10^{120} fine-tuning anywhere. The closure equation that forces the measured capacity exactly is work in progress: the current candidate bridge sits 6.6% from the Λ-located capacity [S5], conditional on the readback map F and the premises CP-1–CP-3. The structural stack delivers Lorentz kinematics, the Einstein branch, the Standard Model quotient, three generations, and one Higgs on their declared branches, carried by 111 sorry-free Lean theorems [C6].



The selection chain and the uniqueness endgame

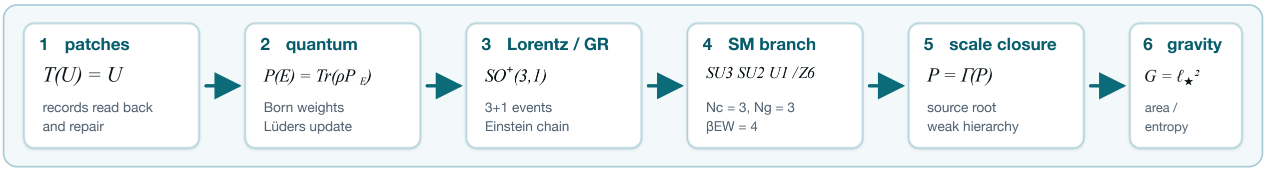
The claim lattice has one logical shape, stated in full in the consistency stack [C15]: ten consistency requirements C1–C10 quotient the space of self-consistent worlds. Self-reading forces records and repair (C1); overlap consensus forces the public quotient, and on the S^2 chart forces signature, dimension, and the Lorentz group (C2); modular self-consistency forces internal time and thermality (C3); transportable-charge consistency forces the compact gauge menu and, with the declared matter package, the Standard Model quotient (C4); entropic consistency forces the Einstein branch and leaves one global number (C5); record existence forces the detuning law and leaves one local number (C6). After C1–C6 the surviving freedom is the two-real-parameter family (P, N) together with the declared discrete selections the ledger counts. C7 identifies the two parameters with measured readouts; C8–C9 are two equations on the family; C10 removes units. Three lemmas close the chain. **L1**: the pixel closure has exactly one fixed point per readout map on the declared physical domain, by the interval and global uniqueness certificates [C1]. **L2**: the same schema applies to N once the readback map F is constructed; its required properties are given by the readback specification [C16]. **L3**: under L1+L2 the principle set admits at most one (P, N); for P this is a theorem on the declared physical domain. If the two equations close and the fixed points are unique, the principle set admits exactly one universe. Whether that universe is ours is the closure ledger reading zero: a theorem-shaped claim whose proof obligation is an experiment, armed and externally timestamped.

Quantitative results beside measurement

Each row carries its evidence class, stated once and in full. Every value is an output of the named declared map or branch, with zero continuous fit parameters.

Quantity	OPH	Reference	Distance	Evidence class
α^{-1}, gauge-width map	137.035660	137.035999177(21) [S2]	2.5×10^{-6} rel.	certified fixed point of the declared map; hadronic transport open
α^{-1}, source map	136.994835	137.035999177(21) [S2]	3.0×10^{-4} rel.	certified fixed point of the declared map; hadronic transport open
m_W chart, pole convention	80.330 GeV	80.3340 GeV converted pole [S3]	$0.5\,\sigma_{\text{exp}}$ (diagnostic)	convention diagnostic on the chart coordinate; physical readout contract open
Capacity at Λ	3.532×10^{122}	3.313×10^{122} [S5]	6.6% (current candidate)	closure equation $N = F(N)$ is work in progress; the exponential hierarchy is structural, the exact bridge is not the final equation
$\Lambda^{(3)}_{\text{QCD}}$	334.8 MeV	338(12) MeV [C7]	$0.3\,\sigma$	conditional source-running branch
$\alpha_s(M_Z)$	0.11834	0.1179(9) [C7]	$0.5\,\sigma$	conditional source-running branch
Scalar tilt n_s	0.9660215	0.9649(42) [S5]	$0.27\,\sigma$	retrospective, trials declared
Cosmic birefringence	0.37501°	$0.342^{+0.094}_{-0.091}{}^{\circ}$ [C7]	$0.35\,\sigma$	retrospective, four trials declared
BTFR slope	4	3.8457(858) [S6]	$1.80\,\sigma$	conditional $\mathbb{Z}_6$ branch, error-aware public-table fit
m_H at measured m_t	125.7 GeV	measured Higgs mass [S4]	within 0.5%, inside declared bands	conditional criticality branch
m_N	929 MeV	938.9 MeV	−1.0%	external lattice input (published m_N/Λ_{QCD} ratio)

The two α rows are landings of the declared maps; the Ward-projected hadronic transport term of the physical Thomson readout is open, so neither row is scored as a physical endpoint verdict. The m_W row is a convention diagnostic in propagated experimental sigma under the complex-pole conversion of the PDG 2026 values [S3]; a physical pull requires the completed readout contract. The capacity row is a work-in-progress lane: the structural result is the exponential hierarchy, and the closure equation that would force the measured capacity exactly is the object under search.



1. The observer-patch fixed point

The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form [O1,O2]. Let T be the observer-overlap repair map and let $\mathfrak{U}_{\text{OPH}}$ denote the selected normal form. The simulation identity is

$$T(\mathfrak{U}_{\text{OPH}}) = \mathfrak{U}_{\text{OPH}}.$$

The OPH simulation theorem establishes this identity from endogenous update, observer-readable records, schedule-independent overlap repair, selected-branch elimination, and implementation/clock closure [O2]. The universe is the stable record world returned by its own readback-and-repair computation.

The concrete carrier is a finite collection of local states, ports, central records, repair maps, and checkpoints. The reference benchmark replays all six overlap schedules and returns the same repaired normal form on the clean abelian branches [C5]. A separate Lean 4/Mathlib artifact certifies observer confluence, approximate schedule independence, refinement naturality, and the contractive conditional-resampling projector, part of the 111 sorry-free theorems that carry the consensus core and the coupling algebra [C6].

Quantum event surface

On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E,

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E), \qquad \rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

These are the Born rule and Lüders update. On the associated two-wing Bell surface,

$$|S_{\text{CHSH}}| \le 2\sqrt{2}.$$

A finite 65,536-record run reproduces the event algebra numerically: committed-event weights sum to 1.0, record projectors have zero idempotency defect, and repeat reads are stable [O1,O3,C7].

2. Relativity, (3+1) events, and Einstein dynamics

The screen readout supplies an oriented conformal S^2. Its connected conformal group is the connected Lorentz group,

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

The corresponding observer-frame space is

$$H^3 \cong \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \qquad \dim H^3 = 6 - 3 = 3.$$

On the declared event branch, repaired records populate a four-dimensional event base of signature $(-+++)$, while H^3 is the fiber of unit timelike observer frames over each event [O4].

The 2π-normalized modular flow of screen caps supplies the observer-relative clock. Half-sided modular inclusions give positive null translations; null tomography assembles their directional charges into a conserved local T_{ab}. Fixed-cap generalized-entropy stationarity, the uniform small-ball limit, and the all-directions tensor upgrade then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}} \langle T_{ab} \rangle.$$

Every arrow in this chain has a named theorem and a machine-readable receipt type [O4,C8].

Earned simulation receipts at 64k, 128k, and 1M patches

Three canonical runs instantiate bounded self-reading patches and scale from 65,536 patches/1,024 observers through 131,072/32,000 to 1,048,576/64,000 [C7]. The 128k and 1M runs earn observer-local modular time, gauge-covariant 2π-KMS replay, a conformal H^3 chart, integer $3+1$ observer-facing experience, and an emergent agreement field; at 1M, 800 tested overlap pairs have median defect zero and cocycle fraction 1.0, against 0.801 for the shuffled control. These are finite structural receipts. Strict neutral bulk, Einstein-branch entry, physical particles, and physical CMB are open lanes in the gap register; the public-data rows above are separately classified paper-side comparisons [C7].

3. From closure coordinates to physical structure

The local pixel closure

The SL-3 estimate of the pixel from measured α is $P = 1.630968209403959\ldots$, a declared input of the claim lattice. A trial pixel P_k produces a field-width readback $R_P(P_k)$, which the pixel chart turns into the next iterate:

$$\alpha_k = R_P(P_k), \qquad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k\sqrt{\pi}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

On the selected particle branch, the internal readback chain is

$$P \longmapsto \alpha_U(P) \longmapsto A_Z(P) \longmapsto A_T^{\text{src}}(P), \qquad \alpha_{\text{in}}(P) = \frac{1}{A_T^{\text{src}}(P)}.$$

Two declared maps carry certified fixed points [C1]. The gauge-width map has the certified self-consistent fixed point

$$\alpha^{-1} = 137.035660136946\ldots,$$

a relative 2.5×10^{-6} from the measured $\alpha^{-1} = 137.035999177(21)$ [S2] (ledger row CL-2). The source map has the certified root

$$\alpha_{\text{root}}^{-1} = 136.994835177413\ldots,$$

a relative 3.0×10^{-4} from measurement (CL-1), with the forward closure point $P_{\text{fwd}} = 1.630972095858897\ldots$ satisfying $P_{\text{fwd}} = \varphi + \sqrt{\pi}/A_T^{\text{src}}(P_{\text{fwd}})$. Both roots are enclosed by direct interval Banach with outward rounding, the enclosure covers the infinite-cutoff SU(2)/SU(3) edge sums through geometric majorants, and the printed-pair identity holds to 35+ digits under converged precision-100 reruns, enforced by a CI test (CL-6). The certificates establish existence and uniqueness for the declared maps; the Ward-projected hadronic transport term of the physical Thomson readout is open, and its source-derived emission under a frozen detached protocol is the registered falsification event for this lane [C18].

The global capacity closure

Let Ω_N^{sc} be the closed-screen normal forms supported at capacity N. The global source map compares their record capacity with the screen budget and applies Minimal Admissible Realization (MAR):

$$S(N) = \log|\Omega_N^{\text{sc}}| - N, \qquad N_\star = \text{MAR}\arg\max_N S(N).$$

The selected endpoint fixes the horizon normalization through

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\text{cell}} = \frac{3\pi P_{\text{fwd}}}{N_\star}, \qquad a_{\text{cell}} = P_{\text{fwd}}\ell_\star^2.$$

The coupling theorem G2-GAP-1 is proven modulo three declared premises: the seed ($N_0 = \pi$), the contraction, and the tick-projection identity are theorems, the algebraic layer is machine-checked in Lean, and the conditional fixed point is certified at 3.532×10^{122} with relative width 1.6×10^{-25} [C16]. The estimate from measured Λ in the Planck base-ΛCDM model is approximately $N_\Lambda = 3.313 \times 10^{122}$ [S5]. The current candidate bridge differs from the Λ-located capacity by 6.6%, so it is not the closure equation that forces the measured capacity; that equation is the object under search (ledger row CL-3). Construction of F and the premises CP-1–CP-3 are open; the balance condition CP-1 is the one counting-theorem-shaped premise between CL-7 and the live capacity test.

Read through $\Lambda_\star \ell_\star^2 = 3\pi/N$, the bridge is a zero-dial relation between the two constants that frame physics:

$$\Lambda\,\ell_P^2 = 3\exp\left[-\frac{6\pi}{P\,\alpha_U(P)}\right] \quad \text{(up to the correction term).}$$

Since α_U is a function of P alone and P is located from α, the theory side holds no continuous freedom once α is measured. The 122 orders of magnitude that the naive vacuum-energy estimate misses appear as e^{-281}, and $281 = 6\pi/(P\,\alpha_U)$ is the same exponent that runs the weak hierarchy: one exponent carries both hierarchy problems. The equation that would force the measured capacity exactly is work in progress; the residual is structural (RG-truncation-shaped terms are excluded by scan). A future Λ determination at percent-level precision tests each candidate closure against the located capacity, so the capacity lane carries a registered way to die that is also a way to win.

The compact-gauge branch

The gauge derivation starts with individually transportable seeds selected by zero-obstruction transport. A common strict representative fixes the tensor category carried by those seeds. Surjective pullback functors and forgetful fibers construct its refinement limit, and Doplicher–Roberts/Tannaka reconstruction reads off the compact group. MAR selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel then fix the hypercharge lattice and quotient [O4]. The result is

$$G_{\mathrm{SM}} = \frac{\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3.$$

The selected hypercharge lattice is

$$\begin{gathered} Q_i = (3,2)_{1/6}, \qquad u_i^c = (\bar{3},1)_{-2/3}, \qquad d_i^c = (\bar{3},1)_{1/3}, \\ L_i = (1,2)_{-1/2}, \qquad e_i^c = (1,1)_1, \qquad H = (1,2)_{1/2}, \qquad i = 1,2,3. \end{gathered}$$

The Higgs doublet is the Borel–Weil carrier

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2, \qquad \mathrm{Stab}(\langle H \rangle) = \mathrm{U}(1)_Q.$$

Its connected adjoint is $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$, which fixes charge quantization and excludes mixed $(3,2,\pm 5/6)$ connected adjoint generators. The same branch supplies the electroweak integer

$$\beta_{\mathrm{EW}} = N_c + 1 = 4.$$

The QCD-free hierarchy witness

The source-audit hierarchy witness reuses the pixel candidate and the compact-gauge integers. Its matched unified-width coordinate is $\alpha_U(P_{\mathrm{fwd}}) = 0.041124247441816685$. The screen count begins with a triangulated S^2. For curvature charge $q_v = 6 - \deg(v)$, Euler's identity and $3F = 2E$ give $\sum_v q_v = 6V - 2E = 12$. Refinement-stable MaxEnt realizes the twelve units as twelve equivalent fivefold defects, and edge-center completion turns them into twelve central ports. Reversible write/check orientation doubles the gauge-algebra dimension: $m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8+3+1) = 24$. With local cell entropy $\ell_{\mathrm{cell}} = P_{\mathrm{fwd}}/4$ and $\beta_{\mathrm{EW}} = 4$, the transmutation law is

$$\frac{v}{E_\star} = P_{\mathrm{fwd}}^{-1/2} \exp\left[-\frac{2\pi}{\beta_{\mathrm{EW}}\alpha_U(P_{\mathrm{fwd}})}\right] = 2.0198114078576331 \times 10^{-17},$$

evaluated at the certified P_{fwd}. The exact parametric law and the 12/24 counts are theorem outputs; the numerical endpoint has the certificate class source-audit branch witness [O5,C3]. The weak scale is a settled-form readout with naturality defect $\epsilon_H = 0$, $\epsilon_H \in [0,0]$: a seventeen-order hierarchy from a one-line law, with no supersymmetric machinery and no tuning.

The criticality branch and the electroweak chart

The criticality law $\lambda = 0$, $\beta_\lambda = 0$ at one source scale fixes the top Yukawa from the gauge couplings, a family with zero continuous parameters over the boundary scale; the boundary-scale selection is a variational theorem (exactly quadratic record cost on the one-loop chart, capacity-weighted log-mean placement) modulo two finite carrier facts, and the selected boundary is

$$\mu_b = \sqrt{\mu_U E_{\mathrm{cell}}} = E_\star e^{-\pi} P_{\mathrm{fwd}}^{-1/6} = 4.86 \times 10^{17}\ \mathrm{GeV}.$$

Evaluated at the measured top mass, the fit-free criticality curve returns the Higgs coordinate 125.7 GeV at two loops, within 0.5% of the measured value [S4] and inside the declared truncation and matching bands (conditional criticality branch) [C4]. A flow-internal alternative is a certified no-go: $d\beta_\lambda/d\ln\mu$ has no root on $[10^{16}, 1.3 \times 10^{19}]$ GeV, so the boundary scale has to come from the observer-patch side, and the four-candidate boundary-scale registry is frozen before any three-loop computation exists [C4].

The strict source-audit branch also returns the running/tree chart coordinates $(m_W^{\mathrm{chart}}, m_Z^{\mathrm{chart}}) = (80.330,\ 91.119)$ GeV, with no measured mass entering; the physical readout contract (renormalized vev, tadpole scheme, thresholds, matching order, complex-pole map) is open, and no physical pull is defined. One convention diagnostic is on record: under the complex-pole convention with PDG 2026 central masses and widths, the converted W pole is 80.3340 GeV [S3], and the chart W coordinate sits 0.5 propagated experimental standard deviations from it. The diagnostic measures convention distance on the chart coordinate; a physical comparison requires the completed readout contract.

Strong-interaction scale

Dimensional transmutation of the source strong coupling, with no hadronic input, gives $\alpha_s(M_Z) = 0.11834$, 0.5σ from the public reference, and fixes the strong scale

$$\Lambda_{\mathrm{QCD}}^{(3)} = 334.8\ \mathrm{MeV},$$

0.3σ from the measured 338(12) MeV, on the conditional source-running branch. The published lattice ratio $m_N/\Lambda_{\mathrm{QCD}}$ places the nucleon at 929 MeV, -1.0% from measurement, as an external lattice input [C2,C7].

4. Geometric gravity normalization and the SI readout

On the declared Einstein branch, the edge-entropy/area-law theorem identifies the coefficient of the local Einstein equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle$$

and its Newton–Poisson limit with the observer-cell ratio

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}}.$$

The local pixel branch supplies $a_{\mathrm{cell}} = P_{\mathrm{fwd}} \ell_\star^2$ and $\bar{\ell}_{\mathrm{shared}} = P_{\mathrm{fwd}}/4$. Substitution gives

$$\frac{G_{\mathrm{geom}}}{\ell_\star^2} = \frac{P_{\mathrm{fwd}}}{4(P_{\mathrm{fwd}}/4)} = 1, \qquad \boxed{G_{\mathrm{geom}} = \ell_\star^2}.$$

The pixel number cancels because it counts both cell area and shared edge entropy. The factor 4 follows from the ratio between the structural Einstein coefficient and the Newton–Poisson convention [O4].

Newton coupling and the SI clock readout

A cesium clock translates the geometric identity into SI units:

$$\gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c} = \frac{\varepsilon_{\mathrm{Cs}}}{2\pi}, \qquad G_{\mathrm{SI}} = \frac{c^3 \ell_\star^2}{\hbar} = \frac{c^5 \varepsilon_{\mathrm{Cs}}^2}{4\pi^2 \hbar \nu_{\mathrm{Cs}}^2}.$$

For the selected display coordinate $\gamma_\star = 4.955974336548419\ldots \times 10^{-34}$, this gives $G_{\mathrm{SI}} = 6.674299995910528 \times 10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$, beside the 2022 CODATA value $6.67430(15) \times 10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$ [S1]. The agreement is a unit-bookkeeping identity on the selected scale bridge (SL-5): the display coordinate $\gamma_\star$ is chosen on that bridge, so G enters as an input via the bridge. Evidence class: exact geometric normalization plus SL-5 bridge input display [O1,O4,C3].

5. Landmark solution map

The table states the positive result and records its evidence class once. Paper numbers refer to the numbered OPH papers in the References block.

Physics problem	OPH solution on the declared branch	Evidence class	Paper
Observer and measurement problem	An observer is a bounded self-reading record boundary. Compatible local readouts participate in the repair map that returns the public world $T(\mathfrak{U}) = \mathfrak{U}$.	Internal theorem	1,2
Quantum probabilities and update	The finite central record algebra yields the Born rule, Lüders conditioning, and $\lvert S_{\mathrm{CHSH}}\rvert \le 2\sqrt{2}$ on the two-wing event surface.	Internal theorem	1,3
Time, Lorentz symmetry, and three space dimensions	Cap modular flow supplies the relative clock; $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$, and $H^3 \cong \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ has dimension three over a $3+1$ event base.	Receipt-conditional theorem	4
General relativity	Null translations, local stress, generalized-entropy stationarity, and the uniform small-ball limit compose to $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab}\rangle$.	Receipt-conditional theorem	4
Gravity from quantum information	The edge-entropy/area law fixes the Einstein and Newton coefficient: $G_{\mathrm{geom}} = a_{\mathrm{cell}}/(4\bar{\ell}_{\mathrm{shared}}) = \ell_\star^2$.	Internal identity	4
Standard Model gauge and matter structure	DR/Tannaka reconstruction plus MAR, anomaly cancellation, and Yukawa invariance give $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$, the hypercharge lattice, $N_c = N_g = 3$, and one Higgs doublet.	Realized-branch theorem	4
Electroweak breaking and charge quantization	The Borel–Weil Higgs carrier $H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$ has stabilizer $\mathrm{U}(1)_Q$; the connected adjoint fixes the Standard Model charge lattice.	Realized-branch theorem	4
Proton stability	The selected product-group gauge structure contains no grand-unified gauge bosons connecting quarks to leptons and therefore excludes gauge-mediated proton decay on the realized branch.	Realized-branch corollary	1,4
Weak/UV hierarchy and Higgs naturality	The 12-port screen, 24-slot oriented register, P_{fwd}, and $\alpha_U(P_{\mathrm{fwd}})$ emit $v/E_\star \sim 2.02 \times 10^{-17}$ with $\epsilon_H = 0$.	Source-audit witness; closure rows CL-3/CL-4 open	5
Dark energy / cosmological term	Conditional on a constructed readback map F and CP-1–CP-3, the record-capacity branch gives $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$, equivalently $\Lambda \ell_P^2 = 3e^{-6\pi/(P\alpha_U)}$: the 122 missing orders of the vacuum-energy problem as one exponential of the inverse gauge coupling, while vacuum energy lies in the kernel of the local null readout. No 10^{120} fine-tuning anywhere in the theory.	Conditional symbolic closure	1,4
Dark-matter phenomenology	Repair-stress bookkeeping gives the galaxy continuation $\nu_{\mathrm{OPH}}(x) = [1 - e^{-\lambda_{\mathrm{collar}}\sqrt{x}}]^{-1}$ and the deep limit $v^4 = GM_b a_{\mathrm{eff}}$ on the empirically fitted MLS response [S7]; the conditional $\mathbb{Z}_6$ branch has BTFR slope 4, 1.80σ from the error-aware public-table fit [S6].	Empirical-fit continuation	7
Yang–Mills gap	The transfer/repair theorem gives the conditional identity $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}$ under compact-gauge continuum-branch assumptions that contain the Clay-hard content.	Conditional identity	8
Three distinct charged-lepton masses	Stratum theorem on the icosahedral carrier: the fivefold and threefold axes carry exactly doubly degenerate quadrupole spectra, so three distinct masses force the minimizing orbit off the high-symmetry strata; coefficient emission is an open lane.	Closed stratum theorem	5
String-vacuum selection	Observer-patch consistency defines a finite sieve over critical-string candidates; the Bouchard–Donagi geometry is selected on its declared gate stack.	Conditional selection program	9

Where this stands among theory-of-everything candidates

The comparison below is structural and factual; every OPH cell carries a maintained artifact.

	SM + GR	String / M-theory	Loop quantum gravity	OPH
Adjustable parameters	about 26 dials, values unexplained	none in principle, then a landscape of $\sim 10^{500}$ vacua	scope excludes the constants	zero dials; closure-defined constants, discrete selections counted
Constants of nature	inputs	environment-selected on the landscape	outside scope	declared-map landing 2.5×10^{-6} from measured α^{-1}; the 122-order vacuum-energy hierarchy as one exponential of the inverse gauge coupling, with no 10^{120} tuning; exact closure armed
Uniqueness of the solution	n/a	no selection principle	n/a	exactly one fixed point per declared map on the physical domain, certified
Can it die tomorrow	no single kill test	relocates within the landscape	no sharp kill test	33 kill conditions; frozen, timestamped targets; fail-closed gates; permanent executed record
Machine-checked core	no	no standard artifact	partial	111 sorry-free Lean theorems
Hard-problem record	measurement, Λ, hierarchy open	Λ anthropic; hierarchy via SUSY, strained	gravity sector only	measurement, Λ, hierarchy, dark sector, time: addressed structurally, branch conditions on record
Distance to closure	undefined	undefined	undefined	a finite public gap register with per-item statuses [C13]

Freezing pass/fail targets before computing payloads and keeping the executed record on a permanent public ledger have no counterpart in the standard theory-of-everything toolkit. A candidate that cannot be tuned, cannot relocate, and publishes the exact list of ways it can die is playing a different game.

6. The falsification program: how to kill this theory

OPH publishes its own kill conditions. The falsifiability map states exact failure conditions [C14]: 33 kill conditions, cryptographically frozen targets with external timestamp proofs [C18], fail-closed gates on every physical comparison, and a permanent executed record. Targets freeze before payloads are computed, and every executed verdict is kept, whichever way it lands. A full adversarial third-party audit, 42 findings deep, identified no false theorem in the recovered core, and the program absorbed its two implementation defects with fail-closed repairs landed the same day. A reader who wants OPH dead has a published procedure: pick a fork below and produce the object it names. The sharpest standing forks:

Test	Outcome that kills the branch
Gauge-mediated proton decay	Any observation; the realized product-group branch contains no leptoquark gauge boson to mediate it.
Fourth light generation or fourth color	Any discovery; the realized branch fixes $N_g = N_c = 3$.
Charge-lattice outlier	An elementary particle off the derived hypercharge lattice, or a stable fractional-charge color singlet.
Extra light Higgs multiplets or a light superpartner spectrum	Either discovery; the MAR-selected package carries one doublet and no superpartner spectrum.
Source hadronic backend	A source-derived Ward-projected transport function, emitted under a frozen detached protocol, either lands the pixel closure inside its registered basin or kills the payload branch.
Three-loop matching packet	The frozen four-candidate boundary-scale registry either converges onto one candidate or the family is eliminated; the registry predates the computation.
Carrier faithfulness export	The pre-repair observer-patch carrier, exported without tuned coefficients, either matches the frozen equation registry or the criticality branch falls.
Coherence-lift null test	The scoped lift-response continuation fixes the canonical susceptibility; a substrate that reaches the declared coherence contrast with null control channels and no force fraction kills that continuation, with the core reconstruction untouched [C14].
Repair-order countermodel	Two accepted repair orders reaching different normal forms under the stated local-diamond and completeness hypotheses kill the consensus mechanism.
Closure uniqueness	A second solution of the P or N closure, or a demonstrated target leak in a declared source map, kills the compression claim. A second P fixed point anywhere on the declared physical domain is arithmetically excluded by the global uniqueness certificate [C1]; the standing fork moves to domains beyond the declared one.
Capacity closure at future Λ precision	The record-capacity closure equation $N = F(N)$ is work in progress. A Λ determination at percent-level precision tests each candidate closure against the located capacity and kills the candidates that miss; the registration freezes the decision policy in advance [C16].

The open program is finite and enumerated: the hadronic transport (CL-1/CL-2), the capacity readback map F with its contraction certificate (CL-3/CL-4/CL-7), the electroweak physical readout map (CL-5), solver hygiene (CL-6, closed), and a source-derived flavor-orbit selector for the mass sector. Each is a computation or a construction with a declared certificate schema [C16,C17].

The assent structure of this document is deliberately mechanical. A reader who grants the six principles and verifies the cited artifacts retains exactly one degree of freedom: the closure ledger [C17]. The fixed points cannot be relocated, no OPH surface may claim more than the ledger records, and each exit from the argument carries its price in named objects. When the ledger is driven to zero by blind completions with uniqueness certificates, declining the identification of the closure output with our universe requires striking a principle, and each principle names the observation that would justify striking it.

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